

# Unit 1: Quiz - Set 1

**Topics:** Introduction to Data Structures; Array Representation and Operations; Pointers

**Format:** 10 MCQ Questions | **Time:** 5 minutes

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**Q1. What is the primary advantage of using an array over a linked list?**

- a. Dynamic size adjustments
- b. Constant time access to any element given its index
- c. Faster insertion and deletion at the middle
- d. Uses less memory overhead for large datasets

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**Correct Answer: b**

**Q2. The memory address of the first element of an array is commonly referred to as the:**

- a. Peak address
- b. Base address
- c. First pointer
- d. Null address

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**Correct Answer: b**

**Q3. Based on the 2D matrix representation below, how are elements uniquely identified?**

|     |     |     |
|-----|-----|-----|
| 0,0 | 0,1 | 0,2 |
| 1,0 | 1,1 | 1,2 |

Representation of Matrix

- a. By a single sequential index
- b. By a memory address only
- c. By a combination of a row index and a column index
- d. By the value stored inside the cell

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**Correct Answer: c**

**Q4. In C programming, what does the & operator do when placed before a variable?**

- a. Returns the value of the variable
- b. Dereferences the variable
- c. Returns the memory address of the variable
- d. Casts the variable to a pointer

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**Correct Answer: c**

**Q5. Which of the following data structures is considered non-linear?**

- a. Queue
- b. Stack
- c. Linked List
- d. Graph

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**Correct Answer: d**

**Q6. What usually happens if you try to access an array element outside its bounds in C?**

- a. The compiler throws a syntax error
- b. The program automatically resizes the array
- c. It results in undefined behavior or a segmentation fault
- d. It returns a standard NULL value

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**Correct Answer: c**

**Q7. A pointer that has not been initialized to any valid memory address and is currently pointing to a random location is known as a:**

- a. Null pointer
- b. Dangling pointer
- c. Wild pointer
- d. Void pointer

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**Correct Answer: c**

**Q8. How is a 2D array typically laid out in contiguous memory in languages like C?**

Row Major: 

|     |     |     |
|-----|-----|-----|
| 0,0 | 0,1 | 0,2 |
|-----|-----|-----|

Column Major: 

|     |     |     |
|-----|-----|-----|
| 0,0 | 1,0 | 2,0 |
|-----|-----|-----|

Row vs Column Major Order

- a. Diagonal-major order
- b. Column-major order
- c. Random memory allocation
- d. Row-major order

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**Correct Answer: d**

**Q9. Which of the following operations on an unsorted array has a worst-case time complexity of  $O(n)$ ?**

- a. Finding the maximum value
- b. Accessing the 5th element
- c. Updating the first element
- d. Finding the memory size of the array

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**Correct Answer: a**

**Q10. If ptr is a pointer to an integer (and assuming an integer takes 4 bytes), what does ptr++ do?**

- a. Increments the value ptr points to by 1
- b. Advances the pointer's address by 1 byte
- c. Advances the pointer's address by 4 bytes (to the next integer)
- d. Creates a new pointer

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**Correct Answer: c**